

Lecture 28. QR Algorithm without Shifts

The QR algorithm, dating to the early 1960s, is one of the jewels of numerical analysis. Here we show that in its simplest form, this algorithm can be viewed as a stable procedure for computing QR factorizations of the matrix powers $A, A^2, A^3, \dots$.

The QR Algorithm

The most basic version of the QR algorithm seems impossibly simple.

Algorithm 28.1. “Pure” QR Algorithm

$$A^{(0)} = A$$

for $k = 1, 2, \dots$

$$Q^{(k)} R^{(k)} = A^{(k-1)} \quad \text{QR factorization of } A^{(k-1)}$$

$$A^{(k)} = R^{(k)} Q^{(k)} \quad \text{Recombine factors in reverse order}$$

All we do is take a QR factorization, multiply the computed factors Q and R together in the reverse order RQ , and repeat. Yet under suitable assumptions, this simple algorithm converges to a Schur form for the matrix A —upper-triangular if A is arbitrary, diagonal if A is hermitian. Here, to keep the discussion simple, we shall continue to assume as in the last lecture that A is real and symmetric, with real eigenvalues λ_j and orthonormal eigenvectors q_j . Thus our interest is in the convergence of the matrices $A^{(k)}$ to diagonal form.

For convergence to diagonal form to be useful for finding eigenvalues, of course, the operations involved must be similarity transformations. This is easily verified: the QR algorithm first triangularizes $A^{(k)}$ by forming $R^{(k)} = (Q^{(k)})^T A^{(k-1)}$, and the multiplication on the right by $Q^{(k)}$ then gives $A^{(k)} = (Q^{(k)})^T A^{(k-1)} Q^{(k)}$. In fact, we have seen this similarity transformation before: it is the “bad idea” mentioned in Lecture 26. Although this transformation is a bad idea when trying to reduce A to triangular form in a single step, it turns out to be quite powerful as the basis of an iteration.

Like the Rayleigh quotient iteration, the QR algorithm for real symmetric matrices converges cubically. To achieve this performance, however, the algorithm as presented above must be modified by the introduction of shifts at each step. The use of shifts is one of three modifications of Algorithm 28.1 that are required to bring it closer to a practical algorithm:

1. Before starting the iteration, A is reduced to tridiagonal form, as discussed in Lecture 26.
2. Instead of $A^{(k)}$, a shifted matrix $A^{(k)} - \mu^{(k)}I$ is factored at each step, where $\mu^{(k)}$ is some eigenvalue estimate.
3. Whenever possible, and in particular whenever an eigenvalue is found, the problem is “deflated” by breaking $A^{(k)}$ into submatrices.

A QR algorithm incorporating these modifications has the following outline.

Algorithm 28.2. “Practical” QR Algorithm

$(Q^{(0)})^T A^{(0)} Q^{(0)} = A$ for $k = 1, 2, \dots$ Pick a shift $\mu^{(k)}$ $Q^{(k)} R^{(k)} = A^{(k-1)} - \mu^{(k)} I$ $A^{(k)} = R^{(k)} Q^{(k)} + \mu^{(k)} I$ If any off-diagonal element $A_{j,j+1}^{(k)}$ is sufficiently close to zero, set $A_{j,j+1} = A_{j+1,j} = 0$ to obtain $\begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} = A^{(k)}$ and now apply the QR algorithm to A_1 and A_2 .	$A^{(0)}$ is a tridiagonalization of A e.g., choose $\mu^{(k)} = A_{mm}^{(k-1)}$ QR factorization of $A^{(k-1)} - \mu^{(k)} I$ Recombine factors in reverse order
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This algorithm, the QR algorithm with well-chosen shifts, has been the standard method for computing all the eigenvalues of a matrix since the early 1960s. Only in the 1990s has a competitor emerged, the divide-and-conquer algorithm described in Lecture 30.

Tridiagonalization was discussed in Lecture 26, shifts are discussed in the next lecture, and deflation is not discussed further in this book. For now, let

us confine our attention to the “pure” QR algorithm and explain how it finds eigenvalues.

Unnormalized Simultaneous Iteration

Our approach will be to relate the QR algorithm to another method called *simultaneous iteration*, whose behavior is more obvious.

The idea of simultaneous iteration is to apply the power iteration to several vectors at once. (An equivalent term is *block power iteration*.) Suppose we start with a set of n linearly independent vectors $v_1^{(0)}, \dots, v_n^{(0)}$. It seems plausible that just as $A^k v_1^{(0)}$ converges as $k \rightarrow \infty$ (under suitable assumptions) to the eigenvector corresponding to the largest eigenvalue of A in absolute value, the space $\langle A^k v_1^{(0)}, \dots, A^k v_n^{(0)} \rangle$ should converge (again under suitable assumptions) to the space $\langle q_1, \dots, q_n \rangle$ spanned by the eigenvectors $q_1, \dots, q_n$ of A corresponding to the n largest eigenvalues in absolute value.

In matrix notation, we might proceed like this. Define $V^{(0)}$ to be the $m \times n$ initial matrix

$$V^{(0)} = \left[\begin{array}{c|c|c} v_1^{(0)} & \cdots & v_n^{(0)} \end{array} \right], \quad (28.1)$$

and define $V^{(k)}$ to be the result after k applications of A :

$$V^{(k)} = A^k V^{(0)} = \left[\begin{array}{c|c|c} v_1^{(k)} & \cdots & v_n^{(k)} \end{array} \right]. \quad (28.2)$$

Since our interest is in the column space of $V^{(k)}$, let us extract a well-behaved basis for this space by computing a reduced QR factorization of $V^{(k)}$:

$$\hat{Q}^{(k)} \hat{R}^{(k)} = V^{(k)}. \quad (28.3)$$

Here $\hat{Q}^{(k)}$ and $\hat{R}^{(k)}$ have dimensions $m \times n$ and $n \times n$, respectively. It seems plausible that as $k \rightarrow \infty$, under suitable assumptions, the successive columns of $\hat{Q}^{(k)}$ should converge to the eigenvectors $\pm q_1, \pm q_2, \dots, \pm q_n$.

This expectation can be justified by an analysis analogous to that of the last lecture. If we expand $v_j^{(0)}$ and $v_j^{(k)}$ in the eigenvectors of A , we have

$$\begin{aligned} v_j^{(0)} &= a_{1j} q_1 + \cdots + a_{mj} q_m, \\ v_j^{(k)} &= \lambda_1^k a_{1j} q_1 + \cdots + \lambda_m^k a_{mj} q_m. \end{aligned}$$

As in the last section, simple convergence results will hold provided that two conditions are satisfied. The first assumption we make is that the leading $n+1$ eigenvalues are distinct in absolute value:

$$|\lambda_1| > |\lambda_2| > \cdots > |\lambda_n| > |\lambda_{n+1}| \geq |\lambda_{n+2}| \geq \cdots \geq |\lambda_m|. \quad (28.4)$$

Our second assumption is that the collection of expansion coefficients a_{ij} is in an appropriate sense nonsingular. Define $\hat{Q}$ to be the $m \times n$ matrix whose columns are the eigenvectors $q_1, q_2, \dots, q_n$. (Thus $\hat{Q}$, a matrix of eigenvectors, is entirely different from $\hat{Q}^{(k)}$, a factor in a reduced QR factorization.) We assume the following:

$$\text{All the leading principal submatrices of } \hat{Q}^T V^{(0)} \text{ are nonsingular.} \quad (28.5)$$

By the leading principal submatrices of $\hat{Q}^T V^{(0)}$, we mean its upper-left square submatrices of dimensions $1 \times 1, 2 \times 2, \dots, n \times n$. (The condition (28.5) happens to be equivalent to the condition that $\hat{Q}^T V^{(0)}$ has an LU factorization; see Exercise 20.1.)

Theorem 28.1. *Suppose that the iteration (28.1)–(28.3) is carried out and that assumptions (28.4) and (28.5) are satisfied. Then as $k \rightarrow \infty$, the columns of the matrices $\hat{Q}^{(k)}$ converge linearly to the eigenvectors of A :*

$$\|q_j^{(k)} - \pm q_j\| = O(C^k) \quad (28.6)$$

for each j with $1 \leq j \leq n$, where $C < 1$ is the constant $\max_{1 \leq k \leq n} |\lambda_{k+1}|/|\lambda_k|$. As in the theorems of the last lecture, the $\pm$ sign means that at each step k , one or the other choice of sign is to be taken, and then the indicated bound holds.

Proof. Extend $\hat{Q}$ to a full $m \times m$ orthogonal matrix Q of eigenvectors of A , and let Λ be the corresponding diagonal matrix of eigenvalues; thus $A = Q\Lambda Q^T$. Just as $\hat{Q}$ is the leading $m \times n$ section of Q , define $\hat{\Lambda}$ (still diagonal) to be the leading $n \times n$ section of Λ . Then we have

$$V^{(k)} = A^k V^{(0)} = Q\Lambda^k Q^T V^{(0)} = \hat{Q}\hat{\Lambda}^k \hat{Q}^T V^{(0)} + O(|\lambda_{n+1}|^k)$$

as $k \rightarrow \infty$. If (28.5) holds, then in particular, $\hat{Q}^T V^{(0)}$ is nonsingular, so we can multiply the term $O(|\lambda_{n+1}|^k)$ on the right by $(\hat{Q}^T V^{(0)})^{-1} \hat{Q}^T V^{(0)}$ to transform this equation to

$$V^{(k)} = (\hat{Q}\hat{\Lambda}^k + O(|\lambda_{n+1}|^k)) \hat{Q}^T V^{(0)}.$$

Since $\hat{Q}^T V^{(0)}$ is nonsingular, the column space of this matrix is the same as the column space of

$$\hat{Q}\hat{\Lambda}^k + O(|\lambda_{n+1}|^k).$$

From the form of $\hat{Q}\hat{\Lambda}^k$ and the assumption (28.4), it is clear that this column space converges linearly to that of $\hat{Q}$. This convergence can be quantified, for example, by defining angles between subspaces; we omit the details.

Now in fact, we have assumed that not only is $\hat{Q}^T V^{(0)}$ nonsingular, but so are all of its leading principal submatrices. It follows that the argument above also applies to leading subsets of the columns of $V^{(k)}$ and $\hat{Q}$: the first columns,

the first and second columns, the first and second and third columns, and so on. In each case we conclude that the space spanned by the indicated columns of $V^{(k)}$ converges linearly to the space spanned by the corresponding columns of $\hat{Q}$. From this convergence of all the successive column spaces, together with the definition of the QR factorization (28.3), (28.6) follows. $\square$

Simultaneous Iteration

As $k \rightarrow \infty$, the vectors $v_1^{(k)}, \dots, v_n^{(k)}$ in the algorithm (28.1)–(28.3) all converge to multiples of the same dominant eigenvector q_1 of A . Thus, although the space they span, $\langle v_1^{(k)}, \dots, v_j^{(k)} \rangle$, converges to something useful, these vectors constitute a highly ill-conditioned basis of that space. If we actually carried out simultaneous iteration in floating point arithmetic as just described, the desired information would quickly be lost to rounding errors.

The remedy is simple: one must orthonormalize at each step rather than once and for all. Thus we shall not construct $V^{(k)}$ as defined above, but a different sequence of matrices $Z^{(k)}$ with the same column spaces.

Algorithm 28.3. Simultaneous Iteration

Pick $\hat{Q}^{(0)} \in \mathbb{R}^{m \times n}$ with orthonormal columns.

for $k = 1, 2, \dots$

$$Z = A\hat{Q}^{(k-1)}$$

$$\hat{Q}^{(k)} \hat{R}^{(k)} = Z \quad \text{reduced QR factorization of } Z$$

From the form of this algorithm, it is clear that the column spaces of $\hat{Q}^{(k)}$ and $Z^{(k)}$ are the same, both being equal to the column space of $A^k \hat{Q}^{(0)}$. Thus, mathematically speaking, this new formulation of simultaneous iteration converges under the same conditions as the old one.

Theorem 28.2. *Algorithm 28.3 generates the same matrices $\hat{Q}^{(k)}$ as the iteration (28.1)–(28.3) considered in Theorem 28.1 (assuming that the initial matrices $\hat{Q}^{(0)}$ are the same), and under the same assumptions (28.4) and (28.5), it converges as described in that theorem.*

Simultaneous Iteration $\iff$ QR Algorithm

Now we can explain the QR algorithm. It is equivalent to simultaneous iteration applied to a full set of $n = m$ initial vectors, namely, the identity, $\hat{Q}^{(0)} = I$. Since the matrices $\hat{Q}^{(k)}$ are now square, we are dealing with full QR factorizations and can drop the hats on $\hat{Q}^{(k)}$ and $\hat{R}^{(k)}$. In fact, we shall

replace $\hat{R}^{(k)}$ by $R^{(k)}$ but $\hat{Q}^{(k)}$ by $\underline{Q}^{(k)}$ in order to distinguish the Q matrices of simultaneous iteration from those of the QR algorithm.

Here are the three formulas that define simultaneous iteration with $\underline{Q}^{(0)} = I$, followed by a fourth formula that we shall take as a definition of an $m \times m$ matrix $A^{(k)}$:

Simultaneous Iteration:

$$\underline{Q}^{(0)} = I, \quad (28.7)$$

$$Z = A\underline{Q}^{(k-1)}, \quad (28.8)$$

$$Z = \underline{Q}^{(k)} R^{(k)}, \quad (28.9)$$

$$A^{(k)} = (\underline{Q}^{(k)})^T A \underline{Q}^{(k)}. \quad (28.10)$$

And here are the three formulas that define the pure QR algorithm, followed by a fourth formula that we shall take as a definition of an $m \times m$ matrix $\underline{Q}^{(k)}$:

Unshifted QR Algorithm:

$$A^{(0)} = A, \quad (28.11)$$

$$A^{(k-1)} = Q^{(k)} R^{(k)}, \quad (28.12)$$

$$A^{(k)} = R^{(k)} Q^{(k)}, \quad (28.13)$$

$$\underline{Q}^{(k)} = Q^{(1)} Q^{(2)} \dots Q^{(k)}. \quad (28.14)$$

Additionally, for both algorithms, let us define one further $m \times m$ matrix $\underline{R}^{(k)}$,

$$\underline{R}^{(k)} = R^{(k)} R^{(k-1)} \dots R^{(1)}. \quad (28.15)$$

We can now exhibit the equivalence of these two algorithms.

Theorem 28.3. *The processes (28.7)–(28.10) and (28.11)–(28.14) generate identical sequences of matrices $\underline{R}^{(k)}$, $\underline{Q}^{(k)}$, and $A^{(k)}$, namely, those defined by the QR factorization of the k th power of A ,*

$$A^k = \underline{Q}^{(k)} \underline{R}^{(k)}, \quad (28.16)$$

together with the projection

$$A^{(k)} = (\underline{Q}^{(k)})^T A \underline{Q}^{(k)}. \quad (28.17)$$

Proof. We proceed by induction in k . The base case $k = 0$ is trivial. For both simultaneous iteration and the QR algorithm, equations (28.7)–(28.15) imply $A^0 = \underline{Q}^{(0)} = \underline{R}^{(0)} = I$ and $A^{(0)} = A$, from which (28.16) and (28.17) are immediate.

Consider now the case $k \geq 1$ for simultaneous iteration. Formula (28.17) is valid by virtue of the definition (28.10) (they are identical), so we need only verify (28.16), which can be done as follows:

$$A^k = A \underline{Q}^{(k-1)} \underline{R}^{(k-1)} = \underline{Q}^{(k)} R^{(k)} \underline{R}^{(k-1)} = \underline{Q}^{(k)} \underline{R}^{(k)}.$$

The first equality follows from the inductive hypothesis on (28.16), the second from (28.8) and (28.9), and the third from (28.15).

On the other hand, consider the case $k \geq 1$ for the QR algorithm. We can verify (28.16) by the sequence

$$A^k = A \underline{Q}^{(k-1)} \underline{R}^{(k-1)} = \underline{Q}^{(k-1)} A^{(k-1)} \underline{R}^{(k-1)} = \underline{Q}^{(k)} \underline{R}^{(k)}.$$

The first equality follows from the inductive hypothesis on (28.16), the second from the inductive hypothesis on (28.17), and the third from (28.12), together with (28.14) and (28.15). Finally, we can verify (28.17) by the sequence

$$A^{(k)} = (Q^{(k)})^T A^{(k-1)} Q^{(k)} = (\underline{Q}^{(k)})^T A \underline{Q}^{(k)}.$$

The first equality follows from (28.12) and (28.13), and the second from the inductive hypothesis on (28.17). $\square$

Convergence of the QR Algorithm

All the pieces are in place. We can now say a great deal about the convergence of the unshifted QR algorithm.

First, at the level of qualitative understanding: (28.16) and (28.17) are the key. The first of these explains why the QR algorithm can be expected to find eigenvectors: it constructs orthonormal bases for successive powers A^k . The second explains why the algorithm finds eigenvalues. From (28.17) it follows that the diagonal elements of $A^{(k)}$ are Rayleigh quotients of A corresponding to the columns of $\underline{Q}^{(k)}$ (see Exercise 27.1). As those columns converge to eigenvectors, the Rayleigh quotients converge (twice as fast, by (27.3)) to the corresponding eigenvalues. Meanwhile, (28.17) implies that the off-diagonal elements of $A^{(k)}$ correspond to generalized Rayleigh quotients involving approximations of distinct eigenvectors of A on the left and the right. Since these approximations must become orthogonal as they converge to distinct eigenvectors, the off-diagonal elements of $A^{(k)}$ must converge to zero.

We cannot emphasize too strongly how fundamental equations (28.16) and (28.17) are to an understanding of the unshifted QR algorithm. They are memorable; and from them, everything of importance follows.

As for a more quantitative understanding, we have the following consequence of Theorem 28.2.

Theorem 28.4. *Let the pure QR algorithm (Algorithm 28.1) be applied to a real symmetric matrix A whose eigenvalues satisfy $|\lambda_1| > |\lambda_2| > \cdots > |\lambda_m|$ and whose corresponding eigenvector matrix Q has all nonsingular leading principal submatrices. Then as $k \rightarrow \infty$, $A^{(k)}$ converges linearly with constant $\max_j |\lambda_{j+1}|/|\lambda_j|$ to $\text{diag}(\lambda_1, \dots, \lambda_m)$, and $\underline{Q}^{(k)}$ (with the signs of its columns adjusted as necessary) converges at the same rate to Q .*

Exercises

28.1. What happens if you apply the unshifted QR algorithm to an orthogonal matrix? Figure out the answer, and then explain how it relates to Theorem 28.4.

28.2. The preliminary reduction to tridiagonal form would be of little use if the steps of the QR algorithm did not preserve this structure. Fortunately, they do.

(a) In the QR factorization $A = QR$ of a symmetric tridiagonal matrix A , which entries of R are in general nonzero? Which entries of Q ? (In practice we do not form Q explicitly.)

(b) Show that the tridiagonal structure is recovered when the product RQ is formed.

(c) Explain how Givens rotations or 2×2 Householder reflections can be used in the computation of the QR factorization of a tridiagonal matrix, reducing the operation count far below what would be required for a full matrix.

28.3. A real symmetric matrix A has an eigenvalue 1 of multiplicity 8, while all the rest of the eigenvalues are ≤ 0.1 in absolute value. Describe an algorithm for finding an orthonormal basis of the 8-dimensional eigenspace corresponding to the dominant eigenvalue.

28.4. Consider one step of Algorithm 28.1 applied to a tridiagonal symmetric matrix $A \in \mathbb{R}^{m \times m}$.

(a) If only eigenvalues are desired, then only $A^{(k)}$ is needed at step k , not $Q^{(k)}$. Determine how many flops are required to get from $A^{(k-1)}$ to $A^{(k)}$ using standard methods described in this book.

(b) If all the eigenvectors are desired, then the matrix $\underline{Q}^{(k)} = Q^{(1)}Q^{(2)} \cdots Q^{(k)}$ will need to be accumulated too. Determine how many flops are now required to get from step $k-1$ to step k .